

FF-4DGS-Ego: hand pose, placement, and metric scale

Protocols differ. EgoGrasp, Dyn-HaMR and HaWoR evaluate over 128-frame world-space segments with SLAM at full resolution; ours is feedforward, per 16-frame clip, in the camera frame at 224 px. The numbers below are therefore not a direct head-to-head, and we do not claim to beat these methods on hand metrics.

Table 1. H2O hand pose, PA-MPJPE (mm, lower is better).

Method	PA-MPJPE	Protocol
Dyn-HaMR	16.7	world-space, SLAM (reported)
EgoGrasp	18.9	world-space, SLAM (reported)
HaWoR	30.8	world-space, SLAM (reported)
Ours (FF-4DGS-Ego)	46.8	feedforward, per-clip (clean subj1→4)
WiLoR (single-image)	47.6	our run

The Dyn-HaMR / EgoGrasp / HaWoR H2O PA-MPJPE values are re-runs reported in *EgoGrasp* (arXiv:2601.01050, v2) Table 2; the methods’ own papers do not all self-report H2O PA-MPJPE. **Verify against EgoGrasp v2 Table 2 before submission.**

Table 2. Our H2O numbers (21-joint, mm). PA on the clean subject1→4 split; the absolute columns are from the mixed eval set (clean-split re-eval pending).

	PA-MPJPE	C-MPJPE (abs.)	WRIST (abs.)
Ours (FF-4DGS-Ego)	46.8	58.0 [†]	36.6 [†]

[†] C-MPJPE / WRIST are from the mixed all-subjects eval set, *not* the held-out subject-4 split; a clean-split re-evaluation of the fine-tuned head is pending.

Table 3. Absolute placement gain from the 3D-joint loss (HOT3D, 9-seq split, mm).

	Absolute MPJPE	Hand depth
Shape-only loss (no abs-3D)	81	n/a
With absolute 3D-joint loss (ours)	53 (−35%)	~4.5 cm

Both rows report *absolute* MPJPE (no alignment); the Procrustes-aligned shape error (PA-MPJPE) is ≈ 8 mm and is not what this table measures.

Table 4. Metric scale and depth (lower is better).

Setting	Source	Depth error
Hand anchor (HOT3D)	in-scene MANO hand (ours)	4.5 cm
Hand anchor (HOT3D)	UniDepth-V2 (depth FM)	15.7 cm
GS scene depth (HOI4D)	no supervision	20.5 cm
GS scene depth (HOI4D)	with GT depth (ours, no mask)	13.4 cm
Object depth (HOT3D)	frozen, no metric loss	62 cm
Object depth (HOT3D)	frozen + hand anchor (sep. ckpt)	135 cm (degrades)

The two object-depth rows are *separate* frozen-backbone checkpoints (hand-anchor weight 0.0 vs 0.1); the anchor distorts non-hand-region depth when the backbone is frozen. The HOI4D 13.4 cm is best-so-far from an interrupted run (see Table 6 note).

Table 5. World-space placement, reported competitors (mm, lower is better).

Method	C-MPJPE	WA-MPJPE (short / long)	Dataset
HaWoR	51.8	22.5 / 27.4	HOI4D
Hand3R	42.6	38.0 / 56.7	HOI4D
Ours (FF-4DGS-Ego, SLAM-free)	93.0	21.1 / 46.6	HOI4D

The HaWoR HOI4D row is a re-run reported in *Hand3R Table II* (arXiv:2602.03200), not HaWoR’s own paper (which has no HOI4D / C-MPJPE); Hand3R’s row is its own. C-MPJPE is absolute camera-frame (no alignment); WA-MPJPE aligns the trajectory (short/long = 30/100 frames). Ours: SLAM-free feedforward per-clip chaining (no DROID-SLAM), over 11 HOI4D

sequences (ZY20210800001), one 128-frame segment each; absolute W-MPJPE = 353 mm, and WA-MPJPE (our per-window Umeyama Sim(3)) is 21/47 mm short/long, so the W-vs-WA gap is global chaining/scale drift. For reference, HaWoR reports HOT3D *W-MPJPE* 33.2 / WA 11.3 in its own Table 3 (a different dataset and metric, not directly comparable). Competitors use SLAM for the global trajectory.

Table 6. Scale-mechanism verdicts (trend over training, lower is better).

Route	Quantity	Trend
(b) feedforward scale head	metric-scale residual	14.1 $\rightarrow$ 13.1 cm
(a) partial unfreeze + GT depth	HOT3D object depth	23.8 $\rightarrow$ 22.3 cm
(a) HOI4D dense depth (best-so-far)	dense scene depth	20.5 $\rightarrow$ 13.4 cm

The (a) and (b) trends are directional, from short runs on a few sequences (storage-quota limited); the HOI4D row is best-so-far from a run cancelled at step 80 (residuals still oscillating 3–12 cm), *not* a converged number. They establish that the metric coupling is learnable (b) and that unfreezing pulls the scene toward metric rather than distorting it (a, opposite sign to the frozen-backbone baseline). Converged magnitudes are in progress.